

Automatic Recognition of Quranic Tajweed Rules Across Riwayat: A Corpus and Machine-Learning Study of Idgham and Qalqalah[TODO: confirm final title]

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Abstract. Automatic assessment of Quranic recitation requires systems that determine not only *what* was recited but *how* it was pronounced. We present a riwayat-aware framework for the automatic recognition of Tajweed rules from speech. A corpus of 7,998 verse-level recordings (11.17 h) was built from MP3Quran recitations of Juz' 'Amma by fifteen reciters covering the riwayat of Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'; verse segments were obtained by forced alignment with manual reconciliation, and occurrences of Idgham and Qalqalah were localised through rule-based candidate extraction, manual acoustic annotation and expert verification (705 unique textual occurrences). Recitation segments are described by eight interpretable acoustic descriptors and classified into five Tajweed categories with SVM, random forest, *k*-nearest neighbours, XGBoost and a multilayer perceptron. The proposed representation outperforms an MFCC baseline with deltas for every classifier, reaching a Macro-F1 of 0.6935 with XGBoost and gains of up to +0.1348. Ablation identifies duration and a depth metric as the dominant cues, SMOTE raises Idgham bila-Ghunnah recall from 0.0678 to 0.4915, and cross-riwayat transfer yields Macro-F1 values of 0.6584–0.6718.

Keywords: Quranic recitation , Tajweed recognition , Idgham , Qalqalah , acoustic feature extraction , machine learning , riwayat-aware classification

1 Introduction

Qur'anic recitation is a highly structured form of oral Arabic whose pronunciation is governed by the principles of Tajweed and normally acquired through repeated practice with expert feedback. Conventional assessment therefore depends on expert listening, making frequent, scalable and objective evaluation difficult outside formal instruction. Recitation is further transmitted through recognised *riwayat*—narrations sharing the same Qur'anic text but differing in phonetic, articulatory and recitational detail—so a computational system must treat legitimate riwayat variation as distinct from pronunciation error. This study considers the “Big Three” riwayat: Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'.

Automatic assessment of Qur'anic recitation differs fundamentally from conventional speech recognition, which determines what was recited rather than how particular phonetic sequences were realised; reciter identification models speaker

characteristics, and generic pronunciation assessment provides no explicit rule-level representations. The central challenge is thus to detect specific Tajweed phenomena from the acoustic signal despite speaker variability, riwayat-related differences and scarce annotated resources—a challenge for which appropriately annotated speech data is a prerequisite that existing Qur'anic corpora do not meet (Sect. 2).

Idgham and Qalqalah offer complementary test cases. Idgham is an assimilation between adjacent consonants that modifies the temporal and spectral structure of the affected boundary, optionally with nasalisation (ghunnah); Qalqalah is an echo-like release of the consonants *qaf*, *ta'*, *ba'*, *jim*, *dal* in sukun states. Their contrasting signatures make them suitable for testing whether conventional acoustic descriptors capture rule-specific information while remaining robust across riwayat.

Prior work leaves four aspects insufficiently explored: the dominance of lexical- and speaker-level objectives over explicit Tajweed modelling; the scarcity of studies treating a rule occurrence—rather than an utterance—as the unit of analysis; the absence of rule-level temporal annotations; and the unexamined influence of multiple riwayat on acoustic realisation and detection. This paper therefore treats corpus construction, riwayat identification, rule-level annotation, feature extraction, supervised classification and evaluation as interconnected components of one problem.

The contributions are fourfold. *Resource*: a Qur'anic speech corpus covering the Big Three riwayat, annotated at the Tajweed-rule level. *Method*: a riwayat-aware pipeline connecting annotation with acoustic feature extraction and supervised classification. *Experimental*: a quantitative evaluation of five classifiers on Idgham and Qalqalah across the three riwayat. *Application*: a foundation for computer-assisted recitation systems providing rule-specific, interpretable feedback while respecting legitimate riwayat variation.

2 Related Work

2.1 Qur'anic Speech Resources and Recognition

Existing Qur'anic speech resources differ considerably in scale, speaker diversity, recording conditions, annotation granularity and intended use. Verse-level repositories such as EveryAyah [1] link recordings to verses but provide no temporal Tajweed annotation. The Tarteel dataset [2] contributes about 25,000 verse recordings (67.39 h) from more than 1,200 speakers under realistic conditions, yet its annotations link audio to text rather than to rule realisations. Ar-DAD [3] (15,810 clips from 30 reciters plus 397 imitator recordings) targets speaker identification; QDAT [4] provides expert judgements on three rules from over 150 readers but with narrow coverage and heterogeneous recordings; and a crowdsourced corpus of about 7,000 recitations from 1,287 participants [5] offers only category-level labels (crowd accuracy 0.77, inter-rater agreement 0.63). Large-scale ASR corpora—581,216 ayah recordings (■2,882 h, 44 reciters) [6] and the 263,263-recording Quran-Ayah-Corpus with speaker-disjoint partitions [7]—prioritise transcription

over pronunciation correctness. Transmission-aware resources are more recent: the Annotated Quranic Qira'at Dataset [8] adds qira'at and phonetic-variation metadata to 24,183 recordings from 309 reciters, and Riwaya-ID [9] separates Warsh from Hafs with 82% accuracy using wav2vec 2.0 [10] and Whisper [11]. Collectively these resources trade lexical and speaker coverage against rule-level, temporally localised supervision, so corpus size alone cannot satisfy the requirements of automatic Tajweed recognition.

2.2 Automatic Tajweed Recognition

Acoustic Tajweed assessment began with phoneme-duration analysis of Madd and Ghunnah from ten reciters [12], showing temporal cues to be interpretable but rate-sensitive. Qalqalah has received particular attention, from MFCC representations with vector quantisation [13] to multilayer perceptrons for Qalqalah Kubra [14]. Smartajweed [15] cast recognition as several binary acoustic problems (Idgham Meem, Ikhfaa Meem, Tafkheem Lam, Tarqeeq Lam) with filter-bank features and radial-basis-function SVMs, reporting 99% validation accuracy under restricted conditions. Convolutional networks with MFCC inputs reached 90.8% validation accuracy for Qalqalah from 3,322 samples of four professional reciters [16], while recurrent models on QDAT reported 77.0%, 59.4% and 66.1% for Al-Madd al-Munfasil, Al-Ghunnah al-Mushaddadah and Al-Ikhfaa [17]. Hybrid pipelines combine ASR and forced alignment with a dedicated classifier: TajweedAI [18] reached 100% accuracy internally but 57.14% externally, showing that controlled-condition performance does not imply robust generalisation. Two conclusions recur: Tajweed phenomena are not equally separable acoustically, and accuracy alone is insufficient under class imbalance. Idgham and Qalqalah nonetheless remain fragmented across datasets, features and protocols, with no unified framework spanning multiple reciters and traditions.

2.3 Research Gap and Positioning

Four gaps define the present work. First, large ASR corpora provide verse-level textual supervision whose objective—transcription—does not encode whether a rule was correctly realised. Second, rule-level temporal annotation identifying *where* a phenomenon occurs in continuous recitation is largely absent, preventing models from learning rule realisations independently of context. Third, rule-specific datasets trade speaker diversity against acoustic control: four professional reciters [16] versus crowdsourced but heterogeneous recordings [4]. Fourth, riwayat is treated as a global classification target [9] rather than as a factor shifting the acoustic realisation of individual rules. The proposed study addresses these gaps by formulating Idgham and Qalqalah recognition as a riwayat-aware, fine-grained acoustic learning problem on a rule-annotated corpus that retains reciter and riwayat identity, bridging large-scale Qur'anic ASR research and rule-specific Tajweed assessment.

3 Methodology

The framework treats automatic Tajweed recognition as a sequence of dependent stages (Fig. 1). Full-surah recordings are aligned and segmented into verse-level clips with explicit reciter and riwayat metadata (Sect. 4); Tajweed occurrences are localised acoustically through a hybrid rule-based, expert-verified annotation protocol (Sect. 5); segments are represented by interpretable acoustic descriptors and analysed for redundancy and imbalance (Sect. 6); and the resulting instances support a five-class supervised task evaluated with complementary classifiers, ablations, balancing strategies, an MFCC baseline, cross-riwayat transfer and bootstrap uncertainty estimation (Sects. 7–9). Three principles govern the pipeline: the rule occurrence, not the verse, is the unit of analysis; riwayat identity is preserved at every stage so that legitimate variation can be separated from error; and every derived artefact remains linked to the original signal and to the canonical text.

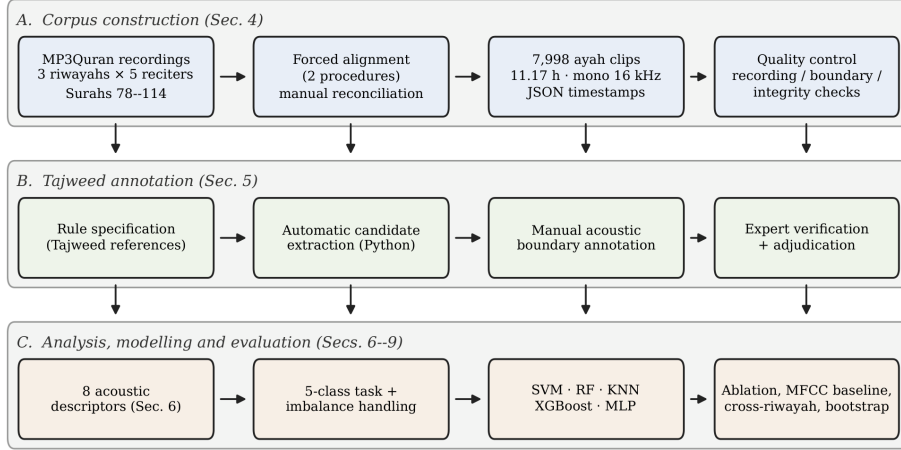


Fig. 1. Overview of the riwayat-aware Tajweed recognition framework: corpus construction with forced alignment and quality control (Sect. 4), hybrid Tajweed annotation (Sect. 5), and acoustic analysis, modelling and evaluation (Sects. 6–9).

4 Corpus Construction

4.1 Source Material and Riwayat Coverage

Audio was collected from the publicly available MP3Quran repository of complete Qur'anic recitations [19]. Rather than restricting the corpus to one tradition, three riwayat were included—Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'—each represented by five reciters, giving fifteen reciters in total. Reciter selection prioritised inter-speaker variability (vocal characteristics, articulation, rate, prosody) within a well-defined tradition, because a model trained on a single reciter may learn

speaker-specific rather than rule-specific characteristics. The scope is Juz' 'Amma (surahs 78–114), i.e. 37 surahs and 564 ayat per reciter, targeting 555 full-surah recordings; the canonical content corresponds to 8,460 reciter–ayah instances, of which approximately 14.14 h of full-surah audio were collected.

4.2 Segmentation, Alignment and Quality Control

Complete surah recordings were processed with two automatic forced-alignment procedures whose timestamps and recognised text were retained separately and reconciled manually into a consolidated segmentation file per surah, from which ayah-level clips were extracted; this prevents alignment or recognition errors from propagating unchecked into the training data. Quality control operated at three levels: recording level (availability, readability, correspondence with the intended surah and reciter); segmentation level (cross-method comparison and review of boundaries); and data-integrity level (expected surah and ayah counts checked against the canonical structure, with missing recordings identified explicitly rather than treated as complete). Alignment confidence scores are stored to flag potentially unreliable segments.

4.3 Corpus Organisation and Statistics

The corpus separates original recordings from derived segments and metadata: per-riwayah directories contain, for each reciter, an `audio/` folder of full-surah recordings, an `ayat/` folder of verse segments, and JSON timestamp files linking every segment to its source position. After segmentation, alignment and validation, 7,998 verse-level recordings were retained—94.5% of the theoretical maximum—comprising 11.17 h of aligned audio distributed evenly across riwayat (Table 1); for modelling, audio is decoded to mono at 16 kHz. The longer mean verse duration of Warsh and Qalun relative to Hafs is consistent with differences in recitation tempo and in the realisation of prolongation (madd) rules.

Table 1. Composition of the verse-level corpus by riwayat.

Riwayah	Reciters	Recordings	Duration (h)	Mean verse dur. (s)
Hafs 'an 'Asim	5	2,739	3.57	4.69
Warsh 'an Nafi'	5	2,562	3.78	5.30
Qalun 'an Nafi'	5	2,697	3.83	5.11
Total	15	7,998	11.17	5.03

Rule extraction from the canonical text identified 328 Idgham and 377 Qalqalah occurrences within Juz' 'Amma—705 unique textual occurrences before replication across reciters and riwayat (Table 2). Because each occurrence is realised by up to fifteen reciters, the number of acoustic instances available for learning is substantially larger than the textual counts, enabling the same phenomenon to be studied under diverse speakers and riwayat. Recording duration is balanced across

riwayat (32–34% each), whereas subtype frequencies are strongly skewed: Lam Shamsiyyah accounts for more than 72% of Idgham occurrences while Idgham Lam-Ra occurs once, and Qalqalah Kubra (12 occurrences) is far rarer than Qalqalah Wusta (145).

Table 2. Unique textual occurrences of the annotated Tajweed subtypes in Juz' 'Amma. Subtypes marked † constitute the five-class recognition task of Sect. 7; the remainder are annotated but excluded from the present classification experiments[**TODO: confirm subtype-exclusion criteria**].

Idgham subtype	Occ.	Qalqalah subtype	Occ.
Lam Shamsiyyah	237	Qalqalah Wusta [†]	145
Idgham bi-Ghunnah [†]	65	Qalqalah Sughra [†]	123
Idgham bila-Ghunnah [†]	25	Qalqalah Mushaddadah [†]	97
Idgham Lam-Ra	1	Qalqalah Kubra	12
Total	328	Total	377

5 Annotation Methodology

Annotation provides rule-specific labels at the level of individual ayat. The protocol derives from established Tajweed references, whose linguistic and phonetic descriptions were translated into explicit, consistently applicable annotation criteria, and combines automatic candidate identification with manual acoustic verification and expert review.

5.1 Idgham

Idgham denotes the assimilation of one consonant into a following consonant, here primarily nun sakinah and tanwin before the letters designated for Idgham, optionally with nasalisation. Acoustically it is expected to remove or reduce an independent realisation of the first consonant, increase temporal overlap between adjacent segments, and yield a boundary dominated by the following consonant, with added low-frequency energy and nasal resonance when ghunnah is present. Candidates were extracted from the Qur'anic text by a Python script encoding the formal Tajweed conditions; the waveform and spectral representation of each corresponding recording were then inspected to localise the realisation and confirm the assimilation. Boundaries were placed to capture the acoustic event while excluding unrelated segments, with particular care where coarticulation, recording quality, rate or context made the boundary ambiguous.

5.2 Qalqalah

Qalqalah is the echo-like release produced by *qaf*, *ta'*, *ba'*, *jim*, *dal* in states requiring the rule, notably sukun states, without an independent vowel. Its signature is a

distinct consonantal release: a transient energy increase, a clear burst, rapid spectral change and a short transition into the neighbouring segment. Candidates were identified textually from the relevant letters and contextual conditions, after which the temporal region of the release was located manually and verified as eligible. Because the realisation varies with reciter, articulation, recording conditions and context, acoustic evidence was always interpreted jointly with the textual condition, and genuine Qalqalah was distinguished deliberately from ordinary consonantal release.

5.3 Workflow, Consistency and Validation

The workflow comprises four steps: *rule specification*, converting Tajweed definitions into annotation criteria; *automatic candidate extraction* from the textual material; *manual acoustic annotation* of temporal boundaries and acoustic parameters; and *expert verification* of rule applicability and segmentation. Consistency was enforced by deterministic textual rules for candidate generation and by predefined boundary criteria applied uniformly. All labels then underwent expert review, with uncertain boundaries, weak realisations and ambiguous conditions examined individually; an annotation was retained only when linguistic condition and acoustic realisation were judged compatible. Inter-annotator agreement was assessed over categorical identification and temporal localisation, with disagreements resolved by expert adjudication[**TODO: report quantitative inter-annotator agreement statistic**]. The ground truth thus combines rule-based extraction, acoustic localisation and expert judgement—essential because a rule's presence in the text does not determine its acoustic realisation.

6 Data Analysis

6.1 Acoustic Feature Representation

Each annotated segment is described by eight handcrafted descriptors chosen to reflect the phonetic mechanisms of the target rules: RMS mean (signal energy), zero-crossing-rate (ZCR) mean (temporal fine structure), spectral-centroid mean (spectral distribution), spectral-flux mean (spectral change), mean and peak nasality (nasal coupling), a depth metric (relative prominence and internal structure of the acoustic event), and duration (temporal extent of the rule-related event). Segment-level statistics are used so that every descriptor is defined uniformly across instances of differing length, and each corresponds to an acoustic dimension that Tajweed theory predicts to be affected by assimilation, nasalisation or consonantal release.

6.2 Redundancy, Normalisation and Cleaning

Correlation analysis indicates partial redundancy: the correlation between ZCR and spectral centroid reaches 0.98, mean and peak nasality correlate at approximately 0.63, and spectral centroid and spectral flux show a moderate positive relationship. Correlated descriptors are not interchangeable, but redundancy motivates normalisation and subset evaluation. Features are standardised as

$$Z_j = \frac{x_j - \mu_j}{\sigma_j}, \quad (1)$$

with μ_j and σ_j estimated from the training partition only, which is essential for distance- and margin-based models[**TODO: confirm the exact normalisation implementation and frame-level extraction parameters**]. A cleaning stage then removed unreliable observations, reducing the instance count in all five classes, most visibly in the majority classes, while Idgham bila-Ghunnah remained substantially under-represented[**TODO: report exact pre-/post-cleaning class counts and cleaning criteria**].

Class distribution. At corpus level the riwayat balance limits riwayat-specific bias, but the subtype distribution is markedly imbalanced: models optimised for overall accuracy may favour dominant classes while failing on rare but linguistically important phenomena, so evaluation must rely on class-sensitive metrics alongside accuracy (Sect. 7.4). The multi-reciter, multi-riwayat design additionally permits testing whether learned representations capture Tajweed-related rather than reciter- or riwayat-specific patterns.

7 Machine Learning Framework

7.1 Task Formulation

Recognition is formulated as supervised multiclass classification: each segmented instance is assigned to one of five Tajweed classes—Idgham bi-Ghunnah, Idgham bila-Ghunnah, Qalqalah Mushaddadah, Qalqalah Sughra and Qalqalah Wusta—from the descriptors of Sect. 6. These five classes are the subtypes of Table 2 retained for the classification experiments; the remaining annotated subtypes lie outside the present experimental scope.

7.2 Classifiers

Five complementary algorithms cover distinct learning paradigms. The soft-margin support vector machine (SVM) minimises $\frac{1}{2}\|\mathbf{w}\|^2 + c\sum \xi_i$ subject to $y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$, combining binary decision functions for the five-class task; it suits compact descriptors but is sensitive to scaling and kernel choice. The random forest (RF) aggregates bootstrap-trained trees over randomised feature subsets by majority vote, modelling nonlinear descriptor interactions and providing importance estimates. k -nearest neighbours (KNN) assigns the majority class among the k closest training observations in feature space, serving as an instance-based

baseline sensitive to local acoustic similarity. XGBoost [20] builds an additive ensemble $\hat{y}_i = \sum f_t(\mathbf{x}_i)$ over regression trees, optimising a regularised objective $\mathcal{L} = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_i)$, which captures higher-order interactions among temporal, spectral and phonatory descriptors while remaining effective for small tabular feature sets. Finally, a multilayer perceptron (MLP) applies hidden-layer transformations $\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$ with a softmax output $P(y = k|\mathbf{x}) = \exp(z_k) / \sum \exp(z_j)$, testing whether nonlinear combinations of the descriptors add discriminative information. Together these models evaluate whether the acoustic representation remains effective across substantially different learning paradigms.

7.3 Class-Imbalance Handling

Imbalance was investigated through five configurations: unbalanced training, class weighting, SMOTE [21], Borderline-SMOTE, and SMOTE with class weighting. SMOTE synthesises minority observations by interpolating between neighbouring minority samples,

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda(\mathbf{x}_{\text{nn}} - \mathbf{x}_i), \lambda \in [0, 1], \quad (2)$$

whereas Borderline-SMOTE restricts synthesis to samples near class boundaries **[TODO: report SMOTE neighbour count and sampling ratio]**. All configurations are evaluated with conventional and imbalance-sensitive metrics, since high overall accuracy can coexist with poor minority-class recognition.

7.4 Evaluation Metrics

Performance is measured by accuracy, per-class precision, recall and F1-score, Macro-F1, weighted F1, balanced accuracy and one-vs-rest ROC-AUC. With $K = 5$ classes, Macro-F1 is the unweighted mean of per-class F1-scores and balanced accuracy the mean of per-class recalls; precision–recall (PR) analysis complements ROC analysis because ROC-AUC can remain high for rare classes even when precision deteriorates at practical operating points.

8 Experiments

8.1 Experimental Protocol

Seven experiment families address the research questions: **E1** evaluates the five classifiers on the cleaned dataset with the proposed representation; **E2** compares it against an MFCC + delta + delta-delta baseline under identical protocols; **E3** analyses per-class behaviour through normalised confusion matrices; **E4** examines discriminability with one-vs-rest ROC and PR curves; **E5** quantifies descriptor contributions through tree-based importance and ablation; **E6** compares the imbalance-handling configurations of Sect. 7.3; and **E7** tests cross-riwayah generalisation by training on one riwayah and evaluating on another for all six

ordered pairs. Uncertainty is estimated by bootstrap resampling of the test set (1,000 resamples).

Protocol details to be finalised. A held-out test set provides final evaluation and 5-fold cross-validation is used for the unbalanced-versus-SMOTE comparison. The train/test proportion, the splitting unit (segment, verse, surah or reciter), stratification and the random seed must be documented to exclude speaker- or recording-level leakage, a particular risk for recitation data[**TODO: specify split strategy, proportion, stratification and seed**]. Hyperparameter grids, selection criteria and final configurations[**TODO: report hyperparameter search and final configurations**], with software versions and hardware[**TODO: report Python/library versions and compute environment**], are likewise required for reproducibility.

9 Results

9.1 Overall Classifier Performance

Table 3 reports the primary comparison (class-weighted configuration). XGBoost achieves the highest Macro-F1 (0.6935) with accuracy 0.7202 and balanced accuracy 0.6953; SVM attains the highest balanced accuracy (0.7126) and random forest the highest accuracy (0.7298), while MLP is weakest on Macro-F1 (0.6219). The divergence between accuracy and Macro-F1—random forest leads on accuracy yet trails XGBoost by 0.0252 on Macro-F1—confirms that accuracy alone mischaracterises this imbalanced problem.

Table 3. Classifier comparison on the held-out test set with the proposed representation (class-weighted configuration). Macro-recall equals balanced accuracy by definition and is reported once.

Model	Accuracy	Balanced acc.	Macro-prec.	Macro-F1	Weighted F1
SVM	0.7083	0.7126	0.6902	0.6920	0.7080
Random forest	0.7298	0.6668	0.7036	0.6683	0.7192
KNN	0.7179	0.6632	0.6872	0.6687	0.7125
XGBoost	0.7202	0.6953	0.6951	0.6935	0.7195
MLP	0.7071	0.6356	0.7001	0.6219	0.6873

9.2 Proposed Representation Versus MFCC Baseline

The proposed descriptors improve Macro-F1 for every classifier (Fig. 2), with the largest gains for random forest (+0.1348), KNN (+0.1336) and XGBoost (+0.1191); accuracy rises from 0.5869 to 0.7298 (random forest), 0.5774 to 0.7179 (KNN) and

0.6119 to 0.7202 (XGBoost). Consistency across all five paradigms indicates that the advantage originates in the information content of the representation rather than in an interaction with a particular decision boundary.

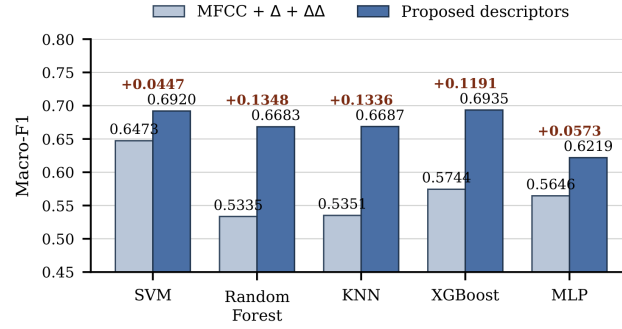


Fig. 2. Macro-F1 of the proposed acoustic descriptors and of the MFCC + delta + delta-delta baseline for each classifier; annotations give the absolute gain of the proposed representation.

9.3 Per-Class Behaviour, Confusion and Discriminability

Per-class results differ sharply (Table 4). Idgham bi-Ghunnah is recognised almost perfectly by every classifier and Qalqalah Wusta remains strong and architecturally stable, whereas Qalqalah Mushaddadah is intermediate and Idgham bila-Ghunnah is the most difficult class; its errors are distributed across several competitors rather than concentrated in one, indicating overlap with multiple categories under the current representation. SVM's relatively high recall for Idgham bila-Ghunnah explains its leading balanced accuracy despite XGBoost's higher Macro-F1. PR analysis shows precision for Idgham bila-Ghunnah falling steeply as recall increases, so its ROC-AUC must not be read as uniform strength.

Table 4. Per-class diagnostics. Recall ranges are taken across the five classifiers; ROC-AUC values are those of XGBoost.

Class	Recall range (5 classifiers)	ROC-AUC (XGBoost)
Idgham bi-Ghunnah	■1.00	1.00
Qalqalah Wusta	0.88–0.92	0.99
Qalqalah Mushaddadah	0.58–0.72	0.89
Idgham bila-Ghunnah	0.017–0.49	0.87
Qalqalah Sughra	0.55 (XGBoost)	0.84

9.4 Feature Importance and Ablation

In XGBoost, duration dominates feature importance (■0.52) followed by the depth metric (■0.21); in random forest the same two descriptors lead (depth ■0.33, duration ■0.24) over a broader distribution. Ablation confirms their causal contribution (Table 5); removing the depth metric reduces Macro-F1 from 0.6935 to

0.5169 (-0.1766) and removing duration to 0.5587 (-0.1348), while depth+duration alone retains 0.6424 against 0.2290 for nasality alone. The full set remains superior, so the remaining descriptors contribute complementary information.

Table 5. Feature ablation with XGBoost on the held-out test set.

Configuration	Accuracy	Balanced acc.	Macro-F1
All features	0.7202	0.6953	0.6935
Remove nasality	0.6964	0.6580	0.6544
Remove duration	0.5798	0.5572	0.5587
Remove depth metric	0.5571	0.5222	0.5169
Remove spectral	0.6905	0.6519	0.6512
Nasality only	0.2393	0.2415	0.2290
Depth + duration only	0.6571	0.6713	0.6424

Class-balancing results. Balancing trades overall accuracy for minority-class sensitivity (Table 6). Unbalanced training yields the highest accuracy (0.7357) but an Idgham bila-Ghunnah recall of only 0.0678 (F1 0.1111). Class weighting raises that recall to 0.3390 and Macro-F1 to 0.6935 at a cost of 1.55 points of accuracy, and SMOTE gives the strongest balanced performance: balanced accuracy 0.6617 ■ 0.7121, Macro-F1 0.6592 ■ 0.6951, and Idgham bila-Ghunnah recall 0.0678 ■ 0.4915 (+42.4 points). Borderline-SMOTE is less effective than standard SMOTE, and adding class weighting after SMOTE yields no further change, indicating that boundary-focused synthesis does not suit this feature space.

Table 6. Effect of class-balancing strategies (XGBoost). “B.-Gh.” denotes the Idgham bila-Ghunnah class.

Configuration	Accuracy	Balanced acc.	Macro-F1	B.-Gh. recall	B.-Gh. F1
Original	0.7357	0.6617	0.6592	0.0678	0.1111
Class weighting	0.7202	0.6953	0.6935	0.3390	0.3478
SMOTE	0.7119	0.7121	0.6951	0.4915	0.3742
Borderline-SMOTE	0.7071	0.6899	0.6812	0.3559	0.3307
SMOTE + weighting	0.7119	0.7121	0.6951	0.4915	0.3742

9.5 Cross-Riwayah Generalisation and Uncertainty

Training on one riwayat and testing on another yields Macro-F1 between 0.6584 and 0.6718 across the six transfer directions (Fig. 3), a narrow range of 0.0134 with the best transfer Qalun ■ Warsh (0.6718) and the weakest Qalun ■ Hafs (0.6584): reasonably stable, yet consistently below the within-distribution XGBoost result, indicating partially riwayat-independent information alongside residual distribution

shift. Bootstrap resampling gives a mean Macro-F1 of 0.6934 with a 95% confidence interval of [0.6604, 0.7240]. McNemar (XGBoost vs. SVM) and Wilcoxon (original vs. SMOTE, 5-fold) tests are specified in the analysis plan but their statistics and p -values are unavailable, so no claim of statistical superiority is made[**TODO: insert McNemar and Wilcoxon statistics and p-values**].

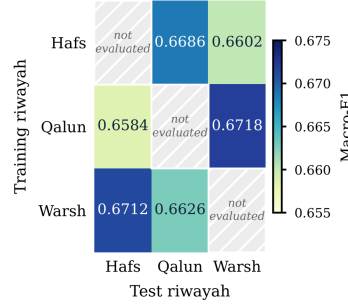


Fig. 3. Cross-riwayah generalisation (Macro-F1): rows are training riwayah and columns testing riwayah. Diagonal cells (within-riwayah evaluation) correspond to the pooled results of Table 3 and are not repeated here.

10 Discussion

10.1 Representation Quality Over Model Complexity

The results indicate that representation quality, rather than classifier complexity, primarily determines performance. MFCCs approximate the short-term spectral envelope and are strong general-purpose descriptors, but Tajweed realisation depends on temporal organisation, articulatory configuration, nasal coupling and consonantal release—dimensions that cepstral coefficients encode only indirectly. That the proposed descriptors improved every classifier, from instance-based KNN to boosted trees and a neural network, supports the interpretation that the gain resides in the acoustic evidence itself. The finding does not establish MFCCs as universally inferior; it shows that, for this task, a compact linguistically motivated descriptor set carries more relevant information than the evaluated MFCC configuration, and that model sophistication cannot compensate for a representation omitting linguistically relevant dimensions.

10.2 Duration and Depth as Acoustic Evidence of Tajweed

Duration and the depth metric converge across two independent forms of evidence—tree-based importance and controlled removal—making an incidental explanation unlikely. This is phonetically plausible: assimilation alters the temporal distribution of energy across a boundary and Qalqalah alters the release behaviour of a consonant, so both produce measurable temporal and structural consequences. More broadly, their prominence supports treating Tajweed as an acoustically

observable realisation of phonetic instructions rather than a purely symbolic property of the text—the premise on which automatic assessment depends. Because duration is also sensitive to annotation boundaries (Sect. 11), its importance should be interpreted jointly with the boundary-uncertainty analysis.

10.3 Why Idgham bila-Ghunnah Remains Difficult

Idgham bi-Ghunnah supplies an explicit nasal cue that nasality descriptors capture directly, whereas Idgham bila-Ghunnah lacks that marker and must rely on subtler temporal and spectral assimilation cues that overlap with neighbouring phonetic events and other target classes. The contrast illustrates the distinction between phonetic definition and acoustic separability: rules unambiguous in theory may partially overlap in measured realisation. Scarcity compounds the problem, and the improvement obtained through SMOTE shows that part of the difficulty is statistical rather than purely acoustic; residual confusion after balancing therefore points to genuine acoustic overlap that no classifier change within the current representation is likely to remove.

10.4 Generalisation Across Riwayat as Domain Shift

The stability of cross-riwayat Macro-F1 suggests that the descriptors capture rule-related information not tied to a single tradition—encouraging for practical assessment, which must accept recitations in different riwayat. The consistent degradation relative to within-distribution performance, however, shows that riwayat-independent classification cannot be assumed: articulation, timing, melodic realisation and recording conditions shift the feature distributions of the same category, particularly for duration- and prosody-sensitive descriptors. Cross-riwayat robustness should therefore be a design objective, with riwayat-related variability treated as a concrete source of domain shift for adaptation or invariant-representation methods.

10.5 Implications for Research and Practice

Two implications follow. Methodologically, Tajweed recognition should be modelled as a structured, rule-aware problem: because categories differ in articulatory mechanism, temporal realisation and acoustic salience, a representation preserving both shared and rule-specific dimensions is preferable to a single undifferentiated vector, and domain knowledge remains valuable even with conventional learners. Practically, interpretable descriptors enable localised, confidence-aware feedback—a suspected rule occurrence, its temporal location and the relevant pronunciation dimension—supporting computer-assisted tutoring beyond global scoring. Such predictions must be presented as assistive feedback rather than definitive religious or pedagogical judgement, with expert supervision retained for difficult cases.

11 Limitations

Corpus scope. The corpus covers Juz' 'Ammah and two rule families only; observed characteristics may not represent the wider Qur'an, and fifteen reciters across three riwayat cannot establish corpus-independent generalisation, so reported gains are evidence within the experimental domain. **Imbalance and synthetic oversampling.** Natural subtype frequencies are skewed, and SMOTE synthesises feature-space samples rather than authentic recitations; gains from oversampling partly reflect better feature-space coverage, not additional phonetic diversity, so class-aware acquisition of genuine minority examples remains necessary. **Annotation uncertainty.** Assimilation phenomena have gradual acoustic transitions, so boundaries are inherently less discrete; because duration and depth depend on boundary placement, small shifts can alter feature values for acoustically equivalent recitations. Quantitative multi-expert agreement analysis and probabilistic boundaries would strengthen reliability assessment. **Acoustic representation.** Segment-level statistics cannot represent long-range dependencies or transitions between neighbouring phonemes, precisely where assimilation evidence may reside; sequential or attention-based architectures should be evaluated alongside—not simply instead of—the interpretable descriptors. **External and statistical validity.** Results derive from one corpus and configuration, so without speaker-independent and external-corpus validation high in-domain performance does not establish robustness to unseen speakers or conditions; and without McNemar and Wilcoxon statistics, superiority claims remain provisional.

12 Conclusion

This paper formulated automatic Tajweed recognition as a riwayat-aware, fine-grained acoustic classification problem and instantiated it end to end: a 7,998-recording, 11.17 h corpus of Juz' 'Ammah spanning Hafsh, Warsh and Qalun; expert-verified rule-level annotation of Idgham and Qalqalah; an interpretable eight-descriptor representation; and a five-class evaluation across five learning paradigms. The representation outperformed an MFCC baseline for every classifier (best Macro-F1 0.6935, XGBoost), ablation established duration and the depth metric as dominant cues, balancing materially improved the difficult Idgham bila-Ghunnah class, and cross-riwayat transfer remained within 0.6584–0.6718. Future work will expand the corpus in surahs, reciters, riwayat and rules; quantify annotation agreement; adopt speaker- and corpus-independent evaluation; and investigate sequential, attention-based and multimodal models combining acoustic evidence with textual and phonological constraints, towards interpretable, confidence-aware recitation support that complements expert instruction.

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